

UFUK BOMBAR

ufuk@bombar.dev · github.com/ubombar · linkedin.com/in/ubombar

Education

- **Sorbonne University – Ph.D. Internet Measurements, EDITE doctoral contract**, 11/2024 – 11/2027
 - Researching large-scale Internet event propagation and measurement methodologies.
- **Sorbonne University – M.Sc. Distributed Computing and Computer Vision**, 09/2022 – 09/2024
 - Awarded SFRI Scholarship from Sorbonne University.
- **Bilkent University – B.Sc. Computer Science and Engineering**, 10/2017 – 01/2022
 - Graduated with Honors, *Summa Cum Laude*.

Experience

- **Software Engineering Intern – Nokia Bell Labs**, 04/2024 – 09/2024
 - Architected Kubernetes integration into an in-house orchestration platform for **edge** and **IoT**, cutting deployment time of edge workloads to under 5 seconds.
 - Developed a **Go**-based provisioning tool reducing cluster setup time from 30 min to a single command.
- **Software Engineer – LIP6 Research Laboratory**, 04/2023 – 04/2024
 - Redesigned core systems of EdgeNet, a suite of custom **Kubernetes** controllers written in **Go**, to achieve production-grade stability, paving the way for Internet measurement research on a global edge-cloud deployment.
- **Computer Vision Engineer – ArgosAI Technology**, 08/2021 – 08/2022
 - Improved runway debris detection with a segmentation/classification model in **PyTorch**, cutting false positives from 50+ to under 4 per day.
 - Integrated **Kubeflow** to orchestrate GPU workloads, replacing manual SSH-based job coordination.
- **Full Stack Software Engineer – Borda Technologies**, 06/2020 – 08/2021
 - Maintained and improved an inventory management system for hospitals built with **ASP.NET Core** and **Flutter**, deployed across 20+ sites.
 - Built smartwatch-based infection tracking system with **AWS Lambda** and **Flutter**; deployed internally during the pandemic.
- **Short-Term Visiting Scholar — University of Mississippi, NCCHE**, 06/2019 – 08/2019
 - Studied **ArcGIS**-based flood simulation tools for the Mississippi River Basin.
 - Implemented a parallel traveling salesman **API** in **Python** for geography-aware navigation on the University campus.

Projects and Activities

- **Measurement Platform Analysis Tool (MPAT)**
 - IMC '25 publication: Designed and implemented a high-performance Internet route survey analysis tool in **Go** and **ClickHouse**, enabling 45 TB data analysis in under 6 hours.
- **SWE Club @ Sorbonne University**
 - Organizing (ongoing) biweekly discussions on emerging trends and best practices in software engineering.
- **Kubernetes Extension for Live Container Migration**
 - Developed a **Kubernetes** controller (**Go**) enabling live container migration with **CRIU**, extending the scheduler for dynamic workload relocation.
- **Pandetect Mask Detection System**
 - Developed a containerized video analysis pipeline in **Electron** using **Docker**, combining **TypeScript** and **Python**.
 - Implemented a streaming protocol in **C** for **ESP32-Cam** to support real-time video analysis.

Summary

- **Spoken Languages:** Turkish (native), English (professional), French (intermediate)
- **Programming Languages:** Go, Python, Bash, C, SQL.
- **Tools:** Kubernetes, ClickHouse, Docker, Git, Linux